

Heated Surface Modification, Channel Orientation and Mass Flow in Rectangular Minichannel Flow Boiling: Comparative Analysis and Multi-Model Machine Learning Validation

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Abstract

This study combines working-fluid-resolved descriptive contrasts of heated-surface condition, spatial orientation and nominal mass-flow level with provenance-safe multi-model machine-learning validation for flow boiling in rectangular minichannels. The integrated archive comprises 64,385,791 reconstructed point-level records from 449 experimental source files spanning six dielectric fluids, smooth and developed surfaces, multiple orientations and several flow levels. The physical analysis is organised around three matched-contrast questions. The matched-cell analysis indicates a strongly fluid-dependent surface response, orientation ratios that are frequently close to parity under comparable conditions, and a more directionally consistent positive response to increasing nominal mass flow in the broader subcooled subsets. These contrasts are treated as descriptive rather than causal because the archive is heterogeneous and not factorially balanced. The predictive analysis compares Histogram Gradient Boosting (HGB), XGBoost, CatBoost and an architecturally independent multilayer perceptron (MLP) under increasingly stringent source-file and fluid-domain validation. On the retained common source-file split, MLP produced the higher Nusselt number R^2 (0.912 versus 0.892), whereas HGB produced the higher heat transfer coefficient R^2 (0.865 versus 0.837). Across five repeated source-file group splits, HGB had the higher mean R^2 for both targets, while leave-one-fluid-out validation was substantially more demanding. Permutation importance was dominated by net heat flux, while the surface identifier and orientation angle had low global incremental importance; these importances are predictive rather than causal. The combined results distinguish conditional experimental contrasts, within-domain new-experiment generalisation and transfer to an unseen working-fluid domain.

Keywords: flow boiling; rectangular minichannel; surface condition; spatial orientation; machine learning; source-file validation; fluid-domain transferability.

Nomenclature

Symbol / acronym	Definition	Unit
D_h	Hydraulic diameter	m
G	Mass velocity	kg m ⁻² s ⁻¹
k_f	Fluid thermal conductivity	W m ⁻¹ K ⁻¹
n	Number of matched cells	-
Nu	Nusselt number	-

Symbol / acronym	Definition	Unit
p	Local absolute pressure	Pa
Pr	Prandtl number	-
q_{net}	Net heat flux transferred to the fluid	W m^{-2}
q_w	Wall heat flux descriptor	W m^{-2}
Re	Reynolds number	-
$T_{f,lin}$	Linearly reconstructed bulk fluid temperature	$^{\circ}\text{C}$
$T_{f,used}$	Reference fluid temperature for target reconstruction	$^{\circ}\text{C}$
$T_{f,in}$	Inlet fluid temperature	$^{\circ}\text{C}$
$T_{f,out}$	Outlet fluid temperature	$^{\circ}\text{C}$
T_{sat}	Local saturation temperature	$^{\circ}\text{C}$
T_{wall}	Local wall temperature	$^{\circ}\text{C}$
x	Axial coordinate	m
α	Local heat transfer coefficient	$\text{W m}^{-2} \text{K}^{-1}$
θ	Nominal spatial orientation angle	deg
ΔT_{cond}	Heated wall conductive temperature correction	K
ΔT_{ws}	Wall-to-saturation temperature difference	K
fluid_id	Working fluid identifier	-
surface_id	Heated surface identifier	-
n_channels	Number of parallel minichannels	-
AI	Artificial intelligence	-
ANN	Artificial neural network	-
CatBoost	Categorical boosting	-
CPU	Central processing unit	-
GPU	Graphics processing unit	-
HGB	Histogram Gradient Boosting	-
LOFO	Leave-One-Fluid-Out validation	-
MAE	Mean absolute error	target unit
MAPE	Mean absolute percentage error	%
ML	Machine learning	-
MLP	Multilayer perceptron	-
PFI	Permutation feature importance	-
QC	Quality control	-
R^2	Coefficient of determination	-
RMSE	Root-mean-square error	target unit
XGBoost	Extreme Gradient Boosting	-

1. Introduction

1.1. Physical background and motivation

Flow boiling in micro- and minichannels is attractive for compact thermal management because combined sensible and latent heat transport can provide high local heat transfer rates with a relatively small working-fluid inventory. The resulting heat transfer behaviour, however, is governed by coupled thermal, hydrodynamic and surface-related phenomena rather than by a single dominant parameter. Fluid properties, confinement, wall heat loading, mass velocity, local pressure, surface morphology and wettability, spatial orientation, bubble dynamics, liquid replenishment and intermittent dry-patch formation may all affect the local wall-to-fluid thermal resistance. Consequently, comparisons performed across heterogeneous experimental conditions may obscure

the conditional character of individual physical effects and confound the influence of surface condition or orientation with differences in working fluid, flow level or boiling regime [1].

Recent experimental and numerical studies reinforce the need for such conditional interpretation. Liu et al. [2] investigated subcooled flow boiling in a rectangular narrow channel and showed that surface modification affected heat transfer through changes in nucleation-site availability, capillary liquid supply and rewetting. Under one of the reported operating conditions, the hybrid surface provided an approximately fourfold enhancement of the heat transfer coefficient and a critical-heat-flux enhancement factor of about 1.62. Vermaak et al. [3] demonstrated that inclination, mass flux and flow instability were coupled in a high-aspect-ratio FC-72 minichannel, with orientation affecting vapour removal and the magnitude of instability-induced wall-temperature excursions rather than acting as an independent correction factor. Ye et al. [4] further showed that engineered microstructures, manifold dimensions and spatial orientation may interact synergistically or antagonistically, depending on the flow orientation. Numerical analysis by Patel et al. [5] similarly linked orientation sensitivity to bubble transit, dry-patch evolution, hydraulic diameter and mass flux. Together, these studies indicate that the thermal consequences of surface modification and spatial orientation depend on their interaction with fluid properties and the relative contributions of buoyancy and inertia.

Earlier investigations conducted within the present experimental research programme showed that spatial orientation, developed surfaces, mass-flow level and other operating conditions interact with working-fluid properties and geometry [6–8,35]. The present study therefore does not reduce these descriptors to background metadata. Instead, it reassesses selected physical parameters through matched descriptive contrasts. Working fluid is treated primarily as a stratification and transfer-domain variable, while surface condition, spatial orientation and nominal mass-flow level are the main contrast variables. Geometry, local heat loading, pressure or saturation state and axial position are recognised as potential confounders; accordingly, the physical contrasts are interpreted descriptively rather than as causal effects.

1.2. Machine learning for flow boiling prediction: state of the art

Qiu et al. [9] provided an early large-scale demonstration of artificial neural networks for saturated flow boiling prediction using 16,953 measurements from 50 sources and 16 working fluids. The model used dimensionless and thermophysical inputs selected to represent the principal heat transfer mechanisms. In addition to a conventional random holdout, complete source databases were removed from training. Performance remained comparatively strong when the working fluid was represented elsewhere in the training set but deteriorated for a previously unseen fluid. These withheld-source

tests established an important distinction between interpolation within a represented domain and transfer to a new fluid domain.

Bard et al. [10] analysed the same consolidated saturated flow boiling database with seven machine-learning families and several feature-selection procedures, including principal-component analysis, Boruta, recursive feature elimination, LASSO and stepwise selection. Their standard 75/25 split preserved fluid proportions. Support-vector regression gave the strongest overall result, with MAPE close to 11% for the full database and about 10.6% for the best unicategorical configuration, below the Bertsch and Kim-Mudawar correlations on the same dataset. Heat flux, vapour-only Froude number and vapour quality were among the influential predictors. The results also showed that model ranking depends on preprocessing, feature selection and split structure, while pooled accuracy alone does not establish transfer to an unseen experiment or fluid.

Qiu et al. [11] subsequently optimised artificial-neural-network inputs and hyperparameters for mini/micro-channel flow boiling heat transfer coefficient prediction. Physically motivated candidates were screened using Pearson correlation and mutual information, and the final model used a compact selected input set. Repeated random-state assessment showed that adding descriptors did not improve performance monotonically. This is pertinent to the present feature set because several reconstructed thermal variables are mutually dependent or target-proximal; predictive value must therefore be considered together with physical meaning, redundancy and the risk of implicitly reconstructing a derived target from its defining variables.

Traverso et al. [12] used Gaussian Process Regression for microchannel heat transfer prediction. From approximately 15,000 raw observations they constructed about 1000 averaged operating cases for R134a and R245fa, using channel length, diameter, mass flux, inlet pressure and heat flux as inputs. Under a random test split, the reported mean absolute error was about 11.5%. When edge regions of the geometric or operating domain were withheld to create an extrapolation test, the error increased to approximately 66.5%, and predictive uncertainty increased in poorly supported regions. The study therefore provides a direct example of the gap between interpolation accuracy and extrapolation beyond the training support.

Suh et al. [13] reviewed AI-assisted phase-change heat transfer and distinguished meta-analysis, physical feature extraction and data-stream analysis as complementary applications. They identified data inconsistency, measurement uncertainty and high-dimensional phase-change dynamics as persistent obstacles and emphasised physics-centred machine learning, in which physical knowledge structures representations, losses or constraints. This perspective motivates evaluating not only predictive error but also which information remains stable enough to support later correlation-informed or physics-guided models.

Loyola-Fuentes et al. [14] proposed a six-stage regression framework separating data pre-treatment, feature selection, splitting philosophy, training and testing, hyperparameter tuning and performance assessment. Although demonstrated for condensation in microfin tubes, the selected machine-learning model was more accurate than the semi-empirical correlation inside the represented domain but degraded markedly in extrapolation. The result reinforces the need to define the intended level of generalisation before interpreting model accuracy.

Bediako and Elbarghthi [15] compared nine machine-learning models for flow boiling heat transfer coefficient prediction at high saturation temperatures using more than 1500 points from seven literature sources and three working fluids. Both dimensional and dimensionless representations were examined. Gradient Boosting, XGBoost and Random Forest were among the strongest models, with reported MAE values below 10% and R^2 values above 0.95 for leading configurations. The study confirms the competitiveness of boosted trees across alternative physical representations, while its pooled validation does not by itself establish transfer to a completely unseen working fluid.

Noh et al. [16] developed an XGBoost model for saturated flow boiling heat transfer coefficient prediction using 11,470 pre-dryout measurements from 41 sources and 23 working fluids. Post-dryout observations were removed with a dedicated incidence model; permutation feature importance and SHAP (Shapley additive explanations) were used for feature selection, and Optuna for hyperparameter optimisation. Seven physically interpretable dimensionless descriptors were retained, including liquid and vapour Prandtl numbers, reduced pressure, boiling number and Froude-related quantities. The resulting XGBoost model achieved a reported mean absolute percentage-type error of approximately 7.18%, outperforming the empirical correlations and alternative ML models considered in that study. These results support inclusion of XGBoost in the present boosted-tree screening while leaving unseen-fluid transfer as a separate validation question.

Shourehdeli and Gholipour [17] examined 24,614 saturated flow boiling points from 120 literature sources spanning mini- and macrochannels. Gradient boosting, XGBoost regression, Random Forest and ANN models were evaluated, with mean errors of roughly 4% for the strongest models. Complete data subsets were then excluded sequentially to probe generalisation beyond the training mixture. Transfer remained reasonable when target conditions were represented by neighbouring regions of the training domain but deteriorated when operating conditions differed substantially. The result highlights representativeness rather than database size alone; the present LOFO analysis applies a stricter version of this principle by withholding an entire working-fluid domain.

Xia et al. [18] examined the accuracy-interpretability trade-off using an Explainable Boosting Machine (EBM) for microchannel flow boiling heat transfer coefficient prediction. Their database comprised 16,440 measurements from 36 publications and 13 working fluids. EBM was benchmarked

against ANN, XGBoost, LightGBM and empirical correlations; it achieved a MAPE of 14.69%, whereas the ANN reached 7.12%. The additive EBM representation nevertheless exposed individual and interaction shape functions with physically plausible tendencies associated with heat flux, mass flux and dryout. A separate 243-point R410A case study yielded an EBM MAPE of 7.90%, illustrating that interpretability and predictive ranking are distinct modelling objectives.

Jung et al. [19] developed uncertainty-aware Gaussian Process Regression for 11,469 pre-dryout points from 41 independent sources. PFI and SHAP were used to reduce the input space, and several covariance kernels were compared. An automatic-relevance-determination exponential kernel gave the best balance between accuracy and uncertainty calibration, with a reported test MAE of 8.45%; on unseen data the error increased to 17.4%. Physical guidance entered through meaningful dimensionless inputs and interpretable kernel length scales rather than governing-equation constraints. This distinction is important when separating a data-only MLP, a correlation-informed network and a governing-equation-constrained physics-informed neural network.

Phan et al. [20] developed correlation-informed neural networks for mini/micro-channel flow boiling using approximately 17,000 measurements and four empirical correlations embedded as guidance terms. The correlation-informed architecture was more robust than a purely data-driven ANN when training data were limited and retained predictive capability for fluids excluded from training. Performance depended on the quality of the embedded correlation: stronger guidance such as the Kim-Mudawar formulation reduced error more effectively, whereas weak guidance imposed a lower performance ceiling. In the present study, the MLP is therefore retained as a data-driven neural baseline rather than presented as a physics-guided model.

Wang and Kharangate [21] analysed uncertainty in 12,366 flow boiling points from 33 studies and four fluids. Monte Carlo dropout represented epistemic uncertainty, heteroscedastic regression represented aleatoric uncertainty, and a K-nearest-neighbours strategy selected supplemental samples near high-uncertainty regions. Adding 50% of points most similar to high-uncertainty samples reduced average epistemic uncertainty from 0.225 to 0.196 and improved MAPE from 21.31% to 14.3%, whereas supplementation in well-supported regions produced little benefit. Adding physically informative inputs such as mass velocity and heat flux also reduced aleatoric uncertainty and lowered MAPE from 31.06% to 10.56%. The results show that database size is not equivalent to information content.

Ayegba et al. [22] combined literature-scale prediction with a regime-specific annular-flow surrogate. Their first-stage database contained 20,891 saturated flow boiling heat transfer coefficient measurements from 83 studies and 22 fluids. Deep neural-network, support-vector, Random-Forest and gradient-boosting models were compared with the Kim-Mudawar correlation using physics-

aligned scalar and dimensionless features, with reported MAPE values of approximately 6.4% for the microchannel validation subset and 4.3% for the minichannel subset. In the second stage, image-derived interfacial-wave features were linked sequentially to wave velocity and frequency, interfacial shear stress and heat transfer coefficient. This combination of broad tabular learning and regime-specific structure motivates retaining a neural branch while keeping it distinct from the validated tabular reference.

1.3. Research gap

The integrated database, harmonisation and quality-control workflow, and the leakage-safe pilot Random-Forest benchmark were reported previously in the foundation study [36]. The present paper uses that established data resource to address different questions: working-fluid-resolved contrasts of surface condition, spatial orientation and nominal mass flow, multi-model HGB–MLP validation, repeated source-file stability, leave-one-fluid-out transferability and permutation feature importance. Taken together, the literature demonstrates two connected gaps. First, the apparent influence of a surface, orientation or flow level can be confounded by unequal coverage of working fluids, geometry, heat loading, pressure and axial position. Second, conventional random or record-level holdouts predominantly measure interpolation inside a mixed domain, while source-file and whole-fluid exclusion test progressively harder forms of generalisation. Relatively few flow boiling studies combine explicitly matched physical contrasts with repeated provenance-group validation and whole-fluid exclusion within one experimental programme. This joint gap motivates the analysis adopted here.

1.4. Objectives and contributions

The study has two linked objectives. The physical objective is to quantify and visualise selected experimental contrasts without treating the heterogeneous archive as a factorially balanced experiment. Surface condition, spatial orientation and nominal mass-flow level are therefore evaluated as working-fluid-resolved matched descriptive contrasts, with the results presented relative to a parity ratio of one. Because geometry, thermal loading, pressure state and axial coverage are not uniformly matched across all experimental subsets, these contrasts are interpreted descriptively rather than causally. The modelling objective is to determine how conclusions about predictive performance change as validation becomes progressively more stringent. The machine-learning programme includes: (i) boosted-tree screening with HGB, XGBoost and CatBoost; (ii) an MLP as an architecturally independent neural comparator; (iii) a common source-file split and five repeated source-file group splits; (iv) sensitivity analysis to source/domain QC exclusions; and (v) leave-one-fluid-out transfer tests with and without fluid identity as a predictor. The purpose is neither a universal heat transfer correlation nor an unconditional ranking of physical factors or algorithms, but a

separation of matched experimental contrasts, within-domain new-experiment generalisation and unseen-fluid transferability.

2. Experimental basis, data reduction and machine-learning methodology

2.1. Experimental facility, test modules and archive scope

The experimental data were acquired during a long-term programme on flow boiling in modular rectangular-minichannel test sections. All experiments incorporated into the archive were conducted on the same experimental facility, using closely related variants of the test section, instrumentation and measurement procedures. The common facility architecture comprised a closed working-fluid loop with circulation and flow-rate control, thermal conditioning, pressure and temperature measurements, electrical Joule heating of the test section, infrared thermography of the externally accessible heated surface and optical observation through a transparent wall. Individual campaigns differed in working fluid, heated-surface condition, channel configuration, nominal mass-flow level and spatial orientation, while preserving the same general measurement concept. The facility architecture and measurement methodology are documented in earlier investigations [6,7,23,37].

Because the archive combines several test-section variants, no single drawing can represent every geometry used in all campaigns.

Figure 1 presents a representative view of the closed-loop experimental facility, the main elements of the rectangular-minichannel test module, and the thermal, hydraulic and optical measurement systems. Part (a) shows the main components of the working-fluid loop, including the test section, circulation pump, heat exchanger (condenser), pressure-regulation module, filter, Coriolis flow meter and deaerator, together with the infrared camera, data-acquisition system and computer used for measurement recording and processing. Part (b) shows the experimental facility in its assembled laboratory configuration. Part (c) presents a representative cross-section of the test section, including the minichannel containing the working fluid, the Haynes-230 heated wall, the transparent glass plate, the plenum and the stainless-steel base and cover. The infrared camera was used to determine the temperature distribution of the externally accessible heated wall. A high-speed camera, not shown in the simplified scheme in part (a), was additionally used in the original experimental campaigns to record two-phase flow structures through the transparent wall; these image data are not used in the present analysis. Pressure sensors, also not shown in the simplified scheme, provided the pressure data used to define the hydraulic operating conditions, while the Coriolis flow meter provided the corresponding flow measurements for each experimental run.

For all test-section configurations, the central minichannel of the parallel-channel group was used as the representative analysis channel. Infrared wall-temperature data were extracted for this channel,

and the corresponding pressure, flow and fluid-temperature measurements recorded during the same experimental run were used consistently in the subsequent data reduction.

The integrated archive used here contains 449 processed experimental source files and 64,385,791 point-level reconstructed records. Its construction, harmonisation, target reconstruction, quality-control logic and pilot Random-Forest modelling are described in the foundation study [36]. Only the subset and coverage information needed to define the present physical contrasts and machine-learning validation domains is repeated here. Each local record remains traceable to its experimental source file, which is used as the fundamental provenance group in the validation.

The original 3M fluorinated fluids are now legacy products following the manufacturer's exit from per- and polyfluoroalkyl substances (PFAS) manufacturing [24]; thermophysical properties used in the experimental reductions were taken from the corresponding manufacturer technical data [25–30].

The hierarchy of the full archive and the analysis-specific subsets used in the present study is summarised in Table 1.

The retained subcooled modelling dataset comprises 443 experiments (source files): 206 for FC-72, 76 for FC-770, 98 for HFE-649, 22 for HFE-7100 and 41 for HFE-7200. The saturated subset is retained mainly as screening context because the initial modelling showed markedly lower and less stable predictive performance for the saturated tasks. The deeper repeated and leave-one-fluid-out validation is therefore focused on the subcooled regime.

The principal experimental ranges extracted from the integrated archive are listed in Table 2. These values describe the physical envelope of the experimental resource rather than the distribution of any one machine-learning subset.

These ranges provide only the principal coverage descriptors required to interpret the contrasts and model domain. The underlying archive is strongly unbalanced across working fluid, surface, orientation and channel configuration; detailed database characterisation is reported in [36].

2.2. Heat transfer quantities and regime definition

The principal response variables are the local heat transfer coefficient α and the Nusselt number Nu . The local thermal reconstruction used net heat flux after the established heat-loss correction, the reconstructed wall temperature and a regime-dependent fluid reference temperature. In compact notation, the target quantities were evaluated as

$$\alpha = q_{net} / (T_{wall} - T_{f,used} - \Delta T_{cond}) \quad (1)$$

$$Nu = \alpha \cdot D_h / k_f \quad (2)$$

where ΔT_{cond} represents the conductive correction through the heated wall and k_f is the fluid thermal conductivity evaluated consistently with the same property-evaluation workflow. The operational

reference temperature $T_{f,used}$ followed the same regime branch throughout the database: the reconstructed bulk-fluid temperature was used for subcooled records and the local saturation temperature for saturated records.

The operational regime classification followed the convention established for the integrated database [36]. The wall-to-saturation temperature difference was defined as $\Delta T_{ws} = T_{wall} - T_{sat}$. Records with $\Delta T_{ws} \leq -2$ K were assigned to the subcooled branch, records with $\Delta T_{ws} \geq +2$ K to the saturated branch, and records within the symmetric deadband $|\Delta T_{ws}| < 2$ K to an ambiguous branch. For the subcooled branch, $T_{f,used} = T_{f,lin}$; for the saturated branch, $T_{f,used} = T_{sat}$. Ambiguous records were excluded from the calculation of α and Nu . These labels are operational data-reduction branches rather than a complete local flow pattern classification.

2.3. Measurement uncertainty

All experiments included in the present archive were conducted on the same experimental facility within a common research programme, using closely related test section variants, instrumentation and measurement procedures. Measurement uncertainties of the recorded quantities and their propagation into local heat transfer calculations were evaluated in the original experimental studies [23]; a detailed uncertainty analysis of the temperature measurement paths used on this research stand is reported in [37]. Because the archive combines several campaigns, working fluids and test section variants, the corresponding source-specific uncertainty estimates were retained as part of the original experimental provenance rather than replaced by a single retrospective archive-wide value. No additional uncertainty-based weighting was applied in the machine-learning training.

2.4. Selected experimental parameters and matched-contrast strategy

Building on the database and QC framework reported in [36], the present analysis examines selected physical parameters across six working fluids (FC-72, FC-770, HFE-649, HFE-7000, HFE-7100 and HFE-7200) and a set of smooth and developed heated surfaces. The developed-surface families include several laser-textured variants, electromachined textures and porous Fe-powder surfaces produced by soldering or sintering; exact source-level identifiers are retained in the metadata. Because individual developed-surface identifiers are sparse in some fluid–orientation–flow combinations, the surface comparison uses a pooled developed-versus-smooth class where matched comparison is supported, while the original surface identifier remains available in the source metadata. The principal contrast variables are heated-surface condition, spatial orientation and nominal mass-flow level. Working fluid defines the thermophysical domain within which those contrasts are interpreted rather than being pooled into a single global effect.

The physical comparisons were constructed from source-file or experiment-level summaries. For this part of the analysis, each source file was assigned a dominant operational regime according to the

largest share of its point-level subcooled, saturated or ambiguous classifications. Surface comparisons retained the same working fluid, dominant regime, nominal orientation and nominal mass-flow group. Orientation comparisons retained the same working fluid, dominant regime, surface class and nominal mass-flow group. Mass-flow comparisons were performed within fixed working-fluid–dominant-regime–surface-class–orientation-class combinations by contrasting the highest and lowest represented nominal flow levels. The response variables were the source-file medians of α and Nu , and the reported ratios were formed within the corresponding matched cells.

These ratios are descriptive contrast measures, not causal estimators. The matching keys do not impose exact equality of geometry, q_{net} , pressure or T_{sat} , heat-loading level, or axial coverage across every comparison; these quantities therefore remain potential confounders. Figures 2–4 use the first-level matched-cell results directly, and no additional fluid–regime synthesis layer is used for the physical results reported in this manuscript. Values close to unity are described as near parity, particularly where the number of matched cells is small.

2.5. Machine-learning dataset, predictors and target transform

The retained subcooled modelling configuration excluded q_w and contained 11 predictors. Nine were numerical: Re , Pr , D_h , q_{net} , T_{sat} , $T_{f,lin}$, $T_{f,used}$, θ and x ; `fluid_id` and `surface_id` were categorical. The wall heat-flux descriptor q_w was not included in the retained configuration, whereas q_{net} remained. The two regression targets were Nu and α . Both targets were modelled after a natural-log transform and predictions were returned to the physical scale using the exponential transform before R^2 , RMSE, MAE, bias and MAPE were calculated. Bias is defined here as the mean prediction error, calculated as the predicted value minus the reference value, so positive values indicate overprediction. In the operational subcooled branch, $T_{f,used}$ equals $T_{f,lin}$ by construction; the two columns therefore contain duplicate information in this branch and their separate contribution is not interpreted as independent physical information.

For the LOFO analysis, `fluid_id` was removed from the predictor list while still being used to define the held-out domain. The remaining predictor set used Re , Pr , D_h , q_{net} , T_{sat} , $T_{f,lin}$, $T_{f,used}$, θ , x and `surface_id`. This prevented an unseen one-hot fluid category from becoming the dominant explanation for transfer failure, but it did not make the representation fluid-complete: surface tension, latent heat, phase-density ratio, vapour viscosity and reduced pressure were not explicit predictors. LOFO is therefore interpreted as transfer under the available feature representation, not as an exhaustive test of fluid-independent boiling physics.

2.6. Multi-model machine-learning framework

The modelling programme was structured hierarchically rather than as a single simultaneous four-model comparison. HGB served as the principal tabular reference because it provided stable

performance at moderate computational cost and represented histogram-based gradient-boosted trees [31]. XGBoost [32] and CatBoost [33] were evaluated in an initial GPU screening stage to determine whether more specialised boosting implementations produced a systematic accuracy gain. The matched GPU results showed CatBoost to be consistently stronger than XGBoost, whereas CatBoost did not systematically outperform HGB. The MLP was therefore retained not as another boosted-tree variant but as an architecturally independent neural comparator for the retained validation.

For the retained HGB configuration, numerical variables were passed directly and `fluid_id` and `surface_id` were one-hot encoded with unknown categories ignored. The `HistGradientBoostingRegressor` used squared-error loss, `learning_rate` = 0.06, `max_iter` = 350, `max_leaf_nodes` = 31, L2 regularisation = 0.01 and `random_state` = 20260602. Early stopping used `validation_fraction` = 0.15 and `n_iter_no_change` = 25. HGB therefore used only the designated training partition for model fitting, with its stopping criterion evaluated on an internal fraction of that training set; the held-out test source files remained untouched until final evaluation.

In the initial GPU screening, XGBoost used `objective` = `reg:squarederror`, 1200 estimators, `learning_rate` = 0.03, `max_depth` = 8, `min_child_weight` = 5, `subsample` = 0.85, `colsample_bytree` = 0.9, `reg_lambda` = 1.0, `reg_alpha` = 0, histogram-based tree construction and CUDA execution. CatBoost used RMSE loss, 1200 iterations, `learning_rate` = 0.03, `depth` = 8, `l2_leaf_reg` = 3.0, `random_seed` = 42 and GPU execution. These screening runs used leakage-safe source-file grouping and source-balanced sampling of up to 3000 rows per source file for training and testing. Their role was model-family screening; their metrics were not merged numerically with the later HGB–MLP validation.

For the final MLP baseline, numerical predictors were standardised using training-partition statistics only. Categorical levels were mapped from the training data and represented by trainable embeddings, with unseen levels assigned to an unknown category. The network comprised three hidden layers of 192 units with SiLU activation and dropout = 0.05. Optimisation used AdamW with learning rate 10^{-3} and weight decay 10^{-4} , mean-squared error in log-target space, `ReduceLROnPlateau` scheduling and early stopping after eight non-improving epochs, with a maximum of 50 epochs. Batch sizes were 8192 for training and 16,384 for validation/test. Unlike HGB, the MLP used the designated source-file validation partition for learning-rate control and early stopping.

2.7. Validation hierarchy

The validation protocol was hierarchical and provenance-safe. First, the GPU boosted-tree screen compared HGB, XGBoost and CatBoost as model families; its numerical results were kept separate from the later HGB–MLP validation. Second, HGB and MLP were evaluated on one fixed common source-file split using the same source-file train/validation/test boundaries and source-balanced

sampling of up to 2000 reconstructed records per source file; the common test partitions contained 175,036 Nu records and 174,361 α records. Third, five repeated source-file group splits were generated using the random seeds 20260602, 20260603, 20260604, 20260605 and 20260606. For each repeat, complete source-file groups were shuffled and 20% were assigned to the test set. Of the remaining groups, 15% were assigned to validation using a second random seed equal to the corresponding split seed plus 1000. Splitting was performed at the source-file level and was not stratified by working fluid. Each repeated dataset was then source-balanced by sampling up to 1000 records per source file using the corresponding split seed, ensuring exact reproducibility of both the group partition and the within-source sampling.

The same group partition was used for HGB and MLP within a split. HGB used an internal fraction of its designated training partition for early stopping, whereas MLP used the designated validation partition. Model configurations were fixed before evaluation on the reported test groups; the held-out test data were not used for hyperparameter selection. The excluded source was one experiment with FC-72, which had been flagged in the preceding source-level QC/high-leverage audit before the retained analysis; the extremely low-coverage subcooled HFE-7000 domain was also excluded consistently. Finally, LOFO removed one complete fluid from training and evaluated only on that unseen fluid, both with and without `fluid_id` as a predictor. This hierarchy separates favourable single-partition performance, new-experiment generalisation, model-ranking stability and new-fluid transferability while preventing related point-level observations from crossing reported source-file test boundaries [34].

3. Descriptive physical contrasts

3.1. Working-fluid-resolved surface-condition contrasts

The first physical comparison contrasts pooled developed and smooth heated surfaces within the subcooled matched subsets. The analysis contains 31 matched cells: 3 for FC-72, 12 for FC-770, 11 for HFE-649 and 5 for HFE-7100. The use of a pooled developed-surface class is a coverage decision rather than an assumption that all textures are equivalent; laser-textured, electromachined and porous surfaces remain distinct in the source metadata, but individual surface families are too sparse for a uniformly resolved comparison across all fluids, orientations and flow levels.

The response is clearly fluid-dependent. FC-72 is below parity for both α and Nu over the matched cells, whereas FC-770 is centred almost exactly on parity. HFE-649 and HFE-7100 have medians above one, but their ranges include or approach parity, so the appropriate interpretation is a conditional positive tendency rather than a universally demonstrated enhancement. In particular, the broad HFE-649 Nu range shows that surface modification can be strongly beneficial in selected conditions without supporting one fluid-independent enhancement factor.

The surface-condition contrasts do not support a single global multiplier. FC-770 is a near-parity case, FC-72 is consistently below parity in the matched cells, and HFE-649/HFE-7100 show positive median responses accompanied by condition dependence. The result therefore supports a working-fluid-resolved interpretation of surface effects.

3.2. Spatial-orientation contrasts

The orientation comparison uses vertical/horizontal ratios within matched cells resolved by working fluid, dominant regime and surface condition. Most FC-72 and FC-770 subcooled median ratios remain close to unity. HFE-649 shows the clearest reduction for vertical operation, especially for Nu on the smooth heated foil surface. For HFE-7100, the response differs between surface conditions: the enhanced-surface subset lies above parity, whereas the smooth-surface subset lies below it. For HFE-7200, the subcooled smooth-surface ratios are also close to parity. These results indicate that the effect of orientation depends on the working fluid and surface condition and cannot be represented by a single universal correction factor.

The limited numbers of matched cells also constrain the interpretation. Several cells contain only one to four matched comparisons, and the only saturated FC-72 smooth-surface subset contains two. Ratios such as 0.96, 1.02 or 1.04 are therefore described as near parity rather than as evidence of a systematic orientation effect. The largest departures from parity occur in selected HFE-649 and HFE-7100 subsets and should be interpreted together with surface condition.

3.3. Nominal mass-flow contrasts

The third contrast compares the highest and lowest represented nominal mass-flow levels within fixed fluid–dominant-regime–surface–orientation subsets. This construction does not assume a linear response with flow rate; it asks only whether the central heat transfer response changes systematically across the covered nominal-flow envelope. In the subcooled subsets, most α and Nu high/low ratios exceed one. The exceptions are the FC-72 smooth inclined subset and the HFE-7100 enhanced horizontal subset, which are at or below parity. The largest subcooled ratios occur for HFE-7200, while HFE-649 also shows a consistent positive response in both smooth and developed subsets. For visual clarity, the saturated subsets are shown separately in Fig. 4(b), because their larger ratios would compress the subcooled range shown in Fig. 4(a).

In the subcooled subsets, nominal mass flow is the most directionally consistent of the three selected contrast variables. The saturated evidence is much less secure: only a few subsets are represented, including very large FC-72 vertical and HFE-649 ratios together with an HFE-7000 near-parity case. These values are therefore treated descriptively and are not used to support a general saturated-flow trend.

3.4. Joint interpretation of the selected physical parameters

Figures 2–4 show that the relative importance of surface condition, orientation and nominal mass flow depends on the working fluid and operating conditions. Surface condition is strongly fluid-dependent; orientation is often a near-parity modifier but can interact with surface morphology; and nominal mass flow provides the most consistent positive direction across the broader subcooled subsets. Working-fluid identity therefore acts as the domain within which the three experimental contrasts must be interpreted.

4. Machine-learning results

4.1. Initial multi-model screening

The ML programme began with regime-specific tabular screening rather than with one preferred algorithm. Table 3 reports the HGB reference performance for four regime-target tasks under the initial source-balanced screening. This stage was used to identify differences in modelling difficulty between regimes and to determine which targets justified the stricter validation hierarchy; it is distinct from the retained common-split and repeated HGB–MLP validation reported later.

The initial screening shows a clear regime difference. The subcooled tasks were substantially more learnable than the saturated tasks, whereas saturated MAPE values exceeded 90% and are unstable in the presence of low target values. Because the foundation study had already identified the same broad regime asymmetry with a pilot Random-Forest benchmark [36], the present screening is retained only to establish the starting point for the subsequent multi-model validation rather than as a separate headline result.

GPU-accelerated XGBoost and CatBoost were evaluated in the same broad screening programme using the HGB feature variants. Table 4 gives the R^2 values obtained after source/domain QC for HGB, CatBoost and XGBoost in the matched screening. CatBoost was the stronger GPU implementation for every task, while XGBoost was materially weaker than CatBoost throughout the four-task comparison.

The boosted-tree comparison yields two separate selection results. First, CatBoost consistently outperformed XGBoost in the matched GPU screening, so XGBoost was not advanced to the final repeated-validation stage. Second, CatBoost did not systematically improve on HGB: its R^2 was lower for saturated Nu , saturated α and subcooled Nu , and only 0.006 higher for subcooled α . The result justified retaining HGB as the tree-based reference while using the MLP to provide an independent neural comparison. Because the boosted-tree screen and the final HGB–MLP validation used distinct sampling configurations and validation stages, the values in Table 4 are not combined into a single

four-model ranking. XGBoost nevertheless remains scientifically relevant because recent flow boiling work reported strong performance for this model family [16].

4.2. Common-split HGB versus MLP

The retained common source-file split was used to compare the tabular HGB reference with the MLP neural baseline under identical source-file train/validation/test boundaries. The documented high-leverage FC-72 source was removed before this analysis, and the very low-coverage subcooled HFE-7000 domain was excluded consistently. Table 5 reports the test metrics; the common test partitions contained 175,036 Nu records and 174,361 α records.

The common source-file split gives a mixed model ranking. For Nu , the MLP produced the higher R^2 (0.912 versus 0.892) and lower RMSE and MAPE. For α , HGB produced the higher R^2 (0.865 versus 0.837), lower RMSE and a mean bias closer to zero. The mixed ranking indicates that model selection based on a single partition is not robust; the ordering must be tested after repartitioning complete experimental source files.

4.3. Repeated source-file-level model stability

Five source-file-level train/validation/test partitions after source/domain QC were used for the HGB–MLP stability analysis. Each test partition contained 89 complete source files, with 300–301 training groups and 53–54 validation groups; sampling was capped at 1000 records per source file. Table 6 reports the mean and between-split standard deviation. Because HGB and MLP used the same provenance partitions within each split, the comparison directly tests model-ranking stability rather than differences in test-set composition.

Across the repeated source-file splits, HGB had the higher mean performance for both targets. For Nu , mean R^2 was 0.794 ± 0.048 for HGB and 0.775 ± 0.040 for MLP; for α , the corresponding values were 0.778 ± 0.054 and 0.749 ± 0.047 . HGB also had lower mean RMSE, MAE and MAPE and a mean bias closer to zero for both targets. In paired partitions, HGB ranked higher in four of five splits for each target. With only five splits, these results support higher mean performance and ranking stability rather than formal statistical superiority.

Because repeated partitions reduce dependence on one particular source-file split, HGB was retained as the tabular reference for the deeper transferability analysis, with MLP retained as the independently structured neural baseline.

4.4. HGB source-file generalisation

For the HGB-specific validation hierarchy, the retained common source-file split is compared directly with the mean of the five repeated source-file splits. The common-split R^2 values were 0.892 for Nu and 0.865 for α , whereas the repeated estimates were 0.794 ± 0.048 for Nu and 0.778 ± 0.054 for α . The repeated results are therefore used to characterise within-domain new-experiment generalisation.

As a sensitivity check, the corresponding HGB analysis before exclusion of the flagged FC-72 source and the very low-coverage subcooled HFE-7000 domain gave common-split R^2 values of 0.896 for Nu and 0.862 for α and repeated mean R^2 values of 0.788 for Nu and 0.774 for α . The small changes after those source/domain QC exclusions argue against a performance-driven gain.

4.5. Leave-one-fluid-out transferability

Source-file-level validation still allows every working fluid to be represented somewhere in training. LOFO therefore provides a stricter test: one complete working fluid is removed from training and used exclusively for testing. The main LOFO analysis removes `fluid_id` from the predictor set so that transfer is based on the available continuous thermohydraulic, thermal, geometric and spatial descriptors together with surface identity. Table 7 reports the per-fluid results. Because the representation omits several fluid-specific properties, these results quantify transfer under the present feature set rather than universal fluid-independent generalisation.

LOFO performance is strongly heterogeneous across working fluids. HFE-7100 transfers well for both targets, and HFE-7200 retains high α R^2 , whereas FC-770 Nu produces a negative R^2 and HFE-649 remains close to zero for both targets. FC-72 shows moderate R^2 but extremely high MAPE because percentage error is unstable at small target values; R^2 , RMSE, MAE and bias therefore carry greater interpretive weight. The key finding is the wide fluid-to-fluid range, shown in Figure 5.

The LOFO results demonstrate a sharp distinction between source-file generalisation and fluid transferability. A model that maintains mean repeated-split R^2 near 0.79 within the known-fluid domain can become ineffective for a specific unseen fluid. This finding is consistent with the deterioration reported in unseen-domain tests by Qiu et al. [9], Traverso et al. [12], Shourehdeli and Gholipour [17] and Jung et al. [19]. It provides a direct experimental justification for treating transferability as a separate modelling endpoint rather than as an assumed consequence of high interpolation accuracy.

4.6. Sensitivity to fluid identity

Because the standard HGB pipeline includes `fluid_id` as a categorical predictor, an additional LOFO sensitivity analysis retained the categorical fluid identifier. Positive ΔR^2 denotes improvement after removing `fluid_id`.

Removing `fluid_id` markedly improved LOFO performance for FC-770 ($\Delta R^2 = +0.422$ for Nu and $+0.346$ for α) and produced smaller changes for the remaining fluids. For HFE-7100 and HFE-7200, removal of `fluid_id` slightly reduced α performance. Retaining `fluid_id` changes the fitted representation, but the held-out category is unseen and is therefore not transferable under this

encoding. Performance remains strongly fluid-dependent after removal of the categorical identifier, consistent with the incomplete fluid-property representation discussed in Section 5.3.

4.7. Permutation feature importance

Permutation feature importance (PFI) was evaluated on the independent test subsets of each of the five repeated HGB splits after source/domain QC. Importance is defined as the mean decrease in physical-scale R^2 after permuting one predictor while leaving the fitted model unchanged. A ΔR^2 value can exceed 1 because R^2 after permutation can become negative. Table 8 reports the mean importance and between-split standard deviation. These values quantify predictive dependence and are not causal effect estimates.

The largest importance is associated with q_{net} for both targets, which is expected because q_{net} is target-proximal in the heat transfer reconstruction. T_{sat} is the second-ranked predictor for both targets. D_h is more influential for Nu , consistent with the definition of Nu , whereas Re has a larger contribution for α . Axial position in the channel x retains non-zero importance, while `surface_id` and θ have small global incremental importance.

PFI should not be read as a causal ranking of physical mechanisms. Correlated predictors can share or mask importance, and target-proximal variables can dominate because they encode part of the reconstructed target. In the subcooled branch, $T_{f,lin}$ and $T_{f,used}$ are identical by construction; therefore, the division of permutation importance between these two columns has no independent physical interpretation. Likewise, the near-zero global PFI of θ and the low PFI of `surface_id` do not contradict the matched physical contrasts in Section 3: PFI asks whether one predictor adds global predictive information across the heterogeneous database, whereas the matched contrasts ask whether conditional physical responses are visible within selected comparable subsets.

5. Discussion

5.1. Physical interpretation of surface, orientation and mass-flow contrasts

The matched comparisons show that the effect of surface modification is strongly fluid-dependent. The enhanced-to-smooth ratios vary between fluids and, in several cases, cross unity; therefore, no universal enhancement factor can be defined. Orientation is frequently a second-order modifier under comparable subcooled conditions, but the opposite HFE-7100 responses for smooth and developed surfaces demonstrate interaction rather than separability. Nominal mass flow is the most directionally consistent comparison in the subcooled subsets, consistent with stronger liquid replenishment and inertial transport at higher flow. These statements remain descriptive because the archive is not factorially balanced and the cell counts are often small.

5.2. Multi-model comparison and model-selection stability

The purpose of the multi-model benchmark was not to identify a universally superior algorithm. HGB, XGBoost and CatBoost represented alternative boosted-tree implementations, whereas the MLP provided a qualitatively different neural architecture. In the GPU screening, CatBoost was consistently stronger than XGBoost but did not systematically surpass HGB. The HGB–MLP common-split comparison did not provide a stable universal ranking: MLP led for Nu , whereas HGB led for α . Across five repeated source-file partitions, HGB had the higher mean R^2 for both targets and ranked higher in four of five splits for each. With only five paired partitions, this evidence supports a statement about higher mean performance and ranking stability, not a claim of statistical superiority. Model-family choice and validation design therefore cannot be separated.

PFI provides a complementary view of the predictors used by the model. The largest predictive dependence is associated with q_{net} , followed by T_{sat} , while the contributions of D_h and Re differ between the two targets. These importance values describe model dependence on the predictors and should not be interpreted as a ranking of physical effects. Therefore, the low global PFI of `surface_id` and θ does not contradict the fluid- and condition-dependent surface and orientation contrasts identified in Section 3. In addition, because $T_{f,used}$ and $T_{f,lin}$ are identical in the subcooled branch, their individual PFI values cannot be given an independent physical interpretation.

5.3. Generalisation versus fluid-domain transferability

The HGB results provide a three-level generalisation hierarchy. The common split shows performance on one reproducible partition; repeated source-file splits reduce dependence on one particular test set; LOFO then exposes a much larger loss of performance for several fluids. These levels answer different engineering questions. A source-file split asks whether the model can predict a new experiment drawn from the known fluid-domain mixture. LOFO asks whether the model can predict a fluid whose complete response surface was absent from training. The negative Nu R^2 for FC-770 and near-zero performance for HFE-649 show that the second question cannot be answered from the first. However, LOFO also probes the adequacy of the chosen feature representation: Re , Pr and T_{sat} do not uniquely encode surface tension, latent heat, phase-density ratio, vapour viscosity or reduced pressure. Part of the transfer loss may therefore reflect omitted fluid-specific physics rather than an intrinsic inability of HGB to extrapolate. Similar degradation under excluded-domain testing has been reported by Qiu et al. [9], Traverso et al. [12], Shourehdeli and Gholipour [17], Jung et al. [19] and Phan et al. [20].

5.4. Neural-network and correlation-informed modelling perspective

The MLP provided the data-driven neural-network baseline and an explicit test of whether an independently structured continuous model would overturn the tree-based conclusion. It achieved the

best Nu result on the retained common source-file split, but that advantage did not persist across repeated source-file partitions; for α it was already weaker than HGB on the common split. The repeated results therefore did not justify replacing the tabular reference with a more complex neural network on the basis of a single favourable partition. Instead, they provide a validated neural baseline for subsequent correlation-informed or physics-guided architectures in which data-driven flexibility is constrained by physically meaningful relationships, for example positivity, consistency between α and Nu , regime-aware activation of boiling terms, or a structured convective-plus-nucleate contribution. The correlation-informed neural-network results of Phan et al. [20] demonstrate that empirical-correlation guidance can improve unseen-fluid robustness, while the review of Suh et al. [13] frames the broader move toward physics-centred ML. Such development lies beyond the scope of the present study.

Classical heat transfer correlations remain relevant as physically structured reference formulations for future correlation-informed models. Their use, however, requires strict adherence to the documented applicability range of each formulation. Because the present archive spans several fluids, regimes and operating domains, a uniform correlation benchmark across the complete dataset would not provide a physically consistent comparison.

5.5. Limitations

The study has several scope limitations. The archive contains a very large number of point-level records but a much smaller number of independent source experiments; validation was therefore performed at the source-file level. Because coverage across working fluid, surface condition, orientation, geometry and operating conditions is unbalanced, the physical results in Figures 2–4 are interpreted as matched descriptive contrasts rather than causal estimates, and cells with few matched comparisons are treated cautiously, particularly when the ratios are close to parity. Fluid-domain coverage is also unequal, so LOFO performance should be interpreted separately for each held-out fluid rather than as a single universal transfer metric. The transfer analysis is further limited to the available predictor set, which does not explicitly include several fluid-specific thermophysical properties; the LOFO results therefore describe transfer under the present feature representation. Finally, the deeper validation focuses on the subcooled regime, while saturated cases are retained mainly as screening context.

6. Conclusions

This study combined working-fluid-resolved descriptive contrasts with a provenance-safe validation hierarchy. The principal conclusions are as follows:

- (1) Surface-condition contrasts are strongly working-fluid-dependent. The FC-72 median ratios lie below parity, FC-770 is close to parity, whereas HFE-649 and HFE-7100 show positive median

tendencies with substantial variation between matched conditions. The results do not support a single universal surface-enhancement factor.

(2) Vertical/horizontal orientation ratios are frequently close to parity in the better represented subcooled subsets. Larger departures occur for selected HFE-649 and HFE-7100 cases and depend on surface condition, indicating that orientation cannot be represented by a single universal correction factor.

(3) Higher nominal mass flow gives the most directionally consistent physical contrast in the subcooled subsets, with most α and Nu high/low ratios above unity. The saturated evidence is substantially sparser and more heterogeneous, so no general conclusion is drawn for that regime.

(4) The model comparison was hierarchical rather than a single simultaneous four-model benchmark. CatBoost consistently outperformed XGBoost in the boosted-tree screening but did not systematically outperform HGB; MLP was therefore retained as an architecturally independent neural comparator.

(5) On the retained common source-file split, MLP produced the higher $Nu R^2$ (0.912 versus 0.892 for HGB), whereas HGB produced the higher αR^2 (0.865 versus 0.837). The mixed ranking demonstrates the sensitivity of model ranking to a single source-file partition.

(6) Across five repeated source-file splits, HGB had the higher mean R^2 for both targets: 0.794 ± 0.048 for Nu and 0.778 ± 0.054 for α , compared with 0.775 ± 0.040 and 0.749 ± 0.047 for MLP. HGB ranked higher in four of five paired partitions for each target, supporting higher mean performance and greater ranking stability without implying formal statistical superiority.

(7) Leave-one-fluid-out validation revealed substantially lower and strongly fluid-dependent transfer performance. Without `fluid_id` as a predictor, $Nu R^2$ ranged from -0.426 for FC-770 to 0.755 for HFE-7100, while αR^2 ranged from 0.081 for HFE-649 to 0.856 for HFE-7100. Because several fluid-specific thermophysical properties were not explicit predictors, these results describe transfer under the available feature representation rather than universal cross-fluid generalisation.

(8) PFI was dominated by q_{net} , followed by T_{sat} , while the contributions of D_h and Re depended on the target. These importance values describe predictive dependence rather than a ranking of physical effects; consequently, the low global importance of `surface_id` and θ does not contradict the conditional surface and orientation contrasts identified in the physical analysis.

(9) The physical and machine-learning analyses therefore provide complementary information: the descriptive contrasts resolve working-fluid- and condition-dependent experimental behaviour, whereas the validation hierarchy distinguishes generalisation to new experiments within represented fluid domains from transfer to an unseen working fluid.

For engineering use, high predictive accuracy within a known mixture of fluid domains should not be interpreted as evidence of cross-fluid transferability. More transferable models should incorporate

a richer representation of fluid-specific thermophysical properties and be validated using complete held-out experimental and fluid domains before cross-fluid generalisation is claimed.

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Data availability

Due to the size and multi-source structure of the complete integrated archive, public deposition of the full dataset is not currently practicable. A representative processed data sample and supporting documentation will be made publicly available in the GitHub repository “boiling-heat-transfer-minichannel” at <https://github.com/MagdalenaPiasecka/boiling-heat-transfer-minichannel> upon publication. The complete experimental datasets, source-file-level metadata, analysis code, and integrated Parquet database are archived in the institutional PRACE-LAB data-storage infrastructure at Kielce University of Technology and are available from the corresponding author upon reasonable request.

CRedit authorship contribution statement

Magdalena Piasecka: Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review and editing.

Declaration of competing interest

The author declares that she has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author used ChatGPT (OpenAI) to assist with language editing, structural organisation and refinement of manuscript wording. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Tables

Table 1. Experimental archive and analysis subsets used in the present study.

Dataset / domain	Source files	Point-level records	Role
Complete integrated archive	449	64,385,791	Provenance base and source for feature reconstruction
Full-archive distribution by fluid	FC-72: 207; FC-770: 76; HFE-649: 98; HFE-7000: 3; HFE-7100: 22; HFE-7200: 43	-	Documents strongly unequal fluid coverage
Retained subcooled ML dataset	443	53,248,518	HGB/MLP validation and leave-one-fluid-out transfer analysis
Retained saturated dataset	440	9,620,349	Initial four-task screening context; not subjected to the same depth of transferability testing

Table 2. Experimental ranges and coverage descriptors represented in the integrated archive.

Parameter	Symbol / unit	Range / coverage represented
Mass velocity	G / kg m ⁻² s ⁻¹	5.4–1375.7
Local absolute channel pressure	p / kPa	90.00–216.76
Wall heat flux	q_w / kW m ⁻²	0.006–104.74
Inlet fluid temperature	$T_{f,in}$ / °C	11.2–28.7
Outlet fluid temperature	$T_{f,out}$ / °C	11.9–30.3
Heated-wall temperature	T_{wall} / °C	13.3–160.2
Nominal spatial orientation	θ / deg	0–270
Working-fluid domains	fluid_id	FC-72; FC-770; HFE-649; HFE-7000; HFE-7100; HFE-7200
Heated-surface conditions	surface_id	Smooth and multiple developed variants, including laser-textured surfaces, electromachined textures and porous Fe-powder surfaces produced by soldering or sintering; source-level surface identifiers retained
Channel-count configuration	n_channels / -	5–25 parallel channels

Table 3. Initial HGB screening results across the four regime-target tasks. For α , RMSE, MAE and bias are in W m⁻² K⁻¹; for Nu they are dimensionless.

Regime	Target	Model	Feature variant	R^2	RMSE	MAE	Bias	MAPE [%]
saturated	Nu	HGB CPU	with q_w	0.488	151.61	104.10	18.31	93.8
saturated	α	HGB CPU	without q_w	0.475	5998.7	4323.8	857.3	91.2
subcooled	Nu	HGB CPU	without q_w	0.857	2.56	1.79	0.16	13.9
subcooled	α	HGB CPU	without q_w	0.784	110.1	78.7	13.4	14.3

Table 4. Broad boosted-tree screening after source/domain QC; ΔR^2 values are relative to the HGB reference.

Regime	Target	HGB R^2	CatBoost GPU R^2	XGBoost GPU R^2	ΔR^2 CatBoost-HGB	ΔR^2 XGBoost-HGB	Interpretation
saturated	Nu	0.488	0.485	0.372	-0.003	-0.116	HGB and CatBoost near parity; XGBoost lower
saturated	α	0.475	0.448	0.324	-0.027	-0.151	HGB best
subcooled	Nu	0.857	0.835	0.748	-0.022	-0.109	HGB best
subcooled	α	0.784	0.790	0.679	+0.006	-0.105	small CatBoost gain

Table 5. Retained common-split HGB and MLP comparison for the subcooled tasks without q_w . For α , RMSE, MAE and bias are in $\text{W m}^{-2} \text{K}^{-1}$; for Nu they are dimensionless.

Target	Model	R^2	RMSE	MAE	Bias	MAPE [%]
Nu	HGB	0.892	2.32	1.52	-0.21	10.9
Nu	MLP	0.912	2.10	1.39	-0.24	9.8
α	HGB	0.865	90.9	63.9	-9.0	10.8
α	MLP	0.837	99.7	72.7	-20.1	11.7

Table 6. Repeated source-file-level HGB–MLP validation; values are mean $\pm$ standard deviation across five splits. For α , RMSE, MAE and bias are in $\text{W m}^{-2} \text{K}^{-1}$; for Nu they are dimensionless.

Target	Model	R^2	RMSE	MAE	Bias	MAPE [%]
Nu	HGB	0.794 $\pm$ 0.048	3.15 $\pm$ 0.44	1.79 $\pm$ 0.16	-0.05 $\pm$ 0.44	12.7 $\pm$ 2.6
Nu	MLP	0.775 $\pm$ 0.040	3.30 $\pm$ 0.38	1.97 $\pm$ 0.14	-0.25 $\pm$ 0.46	13.9 $\pm$ 1.7
α	HGB	0.778 $\pm$ 0.054	118.9 $\pm$ 23.4	69.2 $\pm$ 7.1	-3.2 $\pm$ 14.2	12.1 $\pm$ 1.9
α	MLP	0.749 $\pm$ 0.047	126.7 $\pm$ 21.4	80.8 $\pm$ 4.7	-15.0 $\pm$ 18.5	13.7 $\pm$ 1.6

Table 7. Leave-one-fluid-out HGB results without fluid_id as a predictor. For α , RMSE, MAE and bias are in $\text{W m}^{-2} \text{K}^{-1}$; for Nu they are dimensionless.

Target	Held-out fluid	Test source files	R^2	RMSE	MAE	Bias	MAPE [%]
Nu	FC-72	206	0.458	5.14	3.50	-2.07	87.5
Nu	FC-770	76	-0.426	6.22	5.46	4.90	29.5
Nu	HFE-649	98	0.031	7.06	5.68	4.22	46.9
Nu	HFE-7100	22	0.755	2.88	2.34	2.08	16.3
Nu	HFE-7200	41	0.340	3.96	3.29	3.13	35.1
α	FC-72	206	0.285	220.6	148.5	-117.9	141.7
α	FC-770	76	0.346	155.5	115.1	73.8	17.0
α	HFE-649	98	0.081	233.8	184.9	164.1	41.7
α	HFE-7100	22	0.856	89.7	67.8	46.9	12.5
α	HFE-7200	41	0.751	103.7	82.9	57.1	18.8

Table 8. Permutation feature importance for retained HGB; values are mean $\Delta R^2 \pm$ between-split standard deviation.

Feature	$Nu: \Delta R^2$	$\alpha: \Delta R^2$
$q_{net} [\text{W m}^{-2}]$	1.5790 $\pm$ 0.1957	1.9610 $\pm$ 0.4314
T_{sat}	0.4246 $\pm$ 0.1242	0.4201 $\pm$ 0.1143
D_h	0.2909 $\pm$ 0.0430	0.0324 $\pm$ 0.0105
Re	0.1706 $\pm$ 0.0450	0.2908 $\pm$ 0.0587
x	0.0967 $\pm$ 0.0091	0.1062 $\pm$ 0.0162
fluid_id	0.0392 $\pm$ 0.0202	0.0909 $\pm$ 0.0205
Pr	0.0419 $\pm$ 0.0138	0.0332 $\pm$ 0.0119
T_{film}	0.0380 $\pm$ 0.0143	0.0387 $\pm$ 0.0139
surface_id	0.0076 $\pm$ 0.0042	0.0100 $\pm$ 0.0037
θ	-0.0023 $\pm$ 0.0042	0.0005 $\pm$ 0.0056
$T_{f,used}$	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000

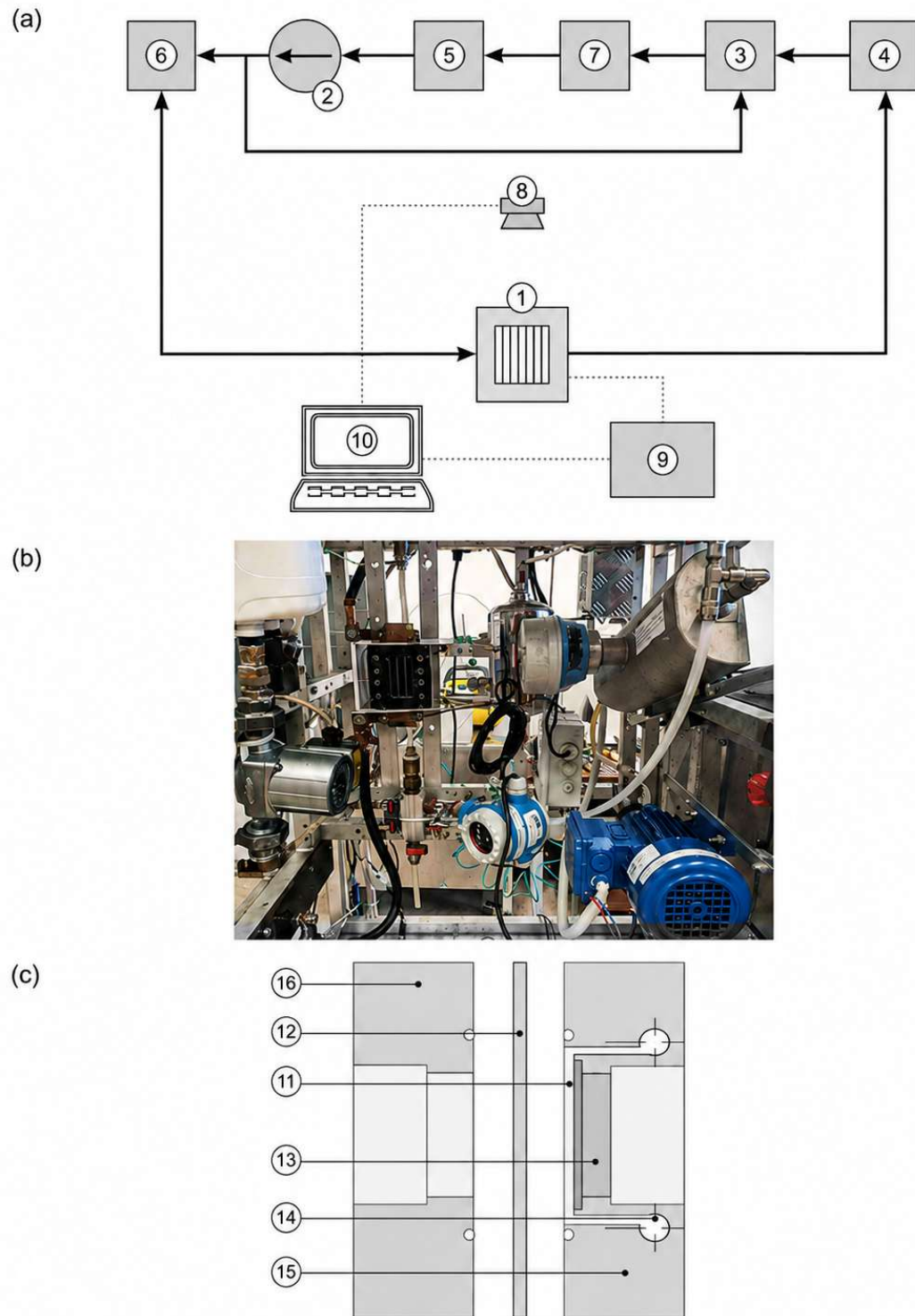


Figure 1. Experimental facility and representative rectangular minichannel test section: (a) main components of the experimental loop and measurement system; (b) photograph of the experimental facility; (c) representative cross-section of the minichannel test section. 1 – test section, 2 – circulation pump, 3 – heat exchanger (condenser), 4 – pressure-regulation module, 5 – filter, 6 – Coriolis flow meter, 7 – deaerator, 8 – infrared (IR) camera, 9 – data-acquisition system, 10 – computer with software, 11 – minichannel with working fluid, 12 – heated wall (Haynes-230 foil), 13 – glass plate, 14 – plenum, 15 and 16 – stainless-steel base and cover.

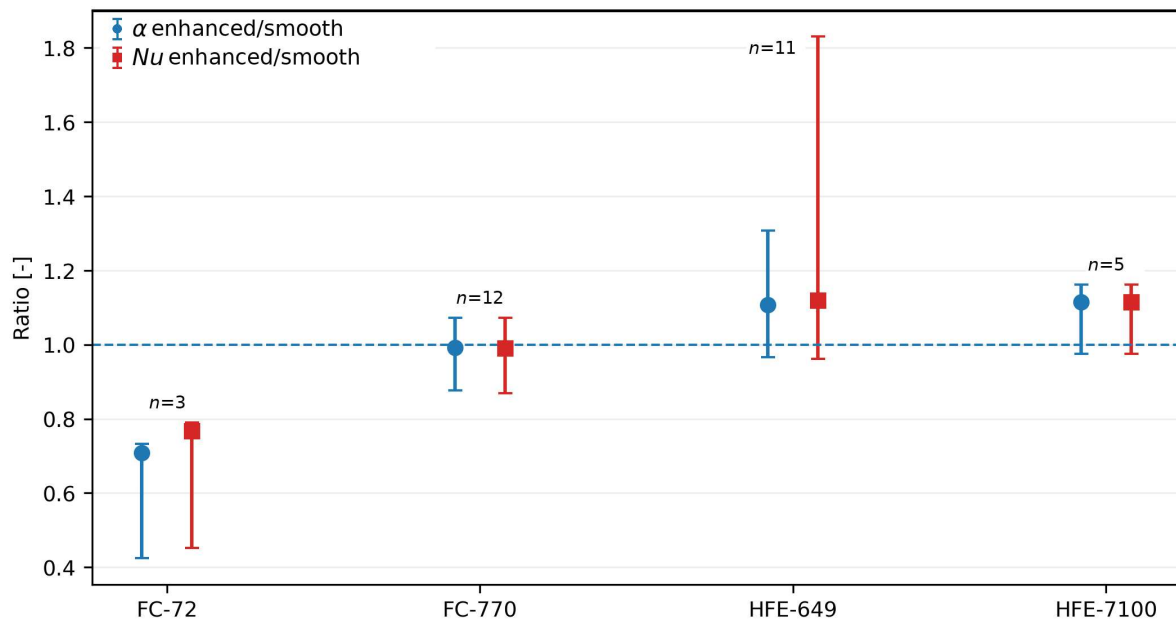


Figure 2. Heated foil surface condition contrasts by working fluid in the subcooled regime. Markers show median enhanced/smooth ratios for α and Nu , whiskers indicate the min–max range, n denotes the number of matched cells, and the dashed line marks parity.

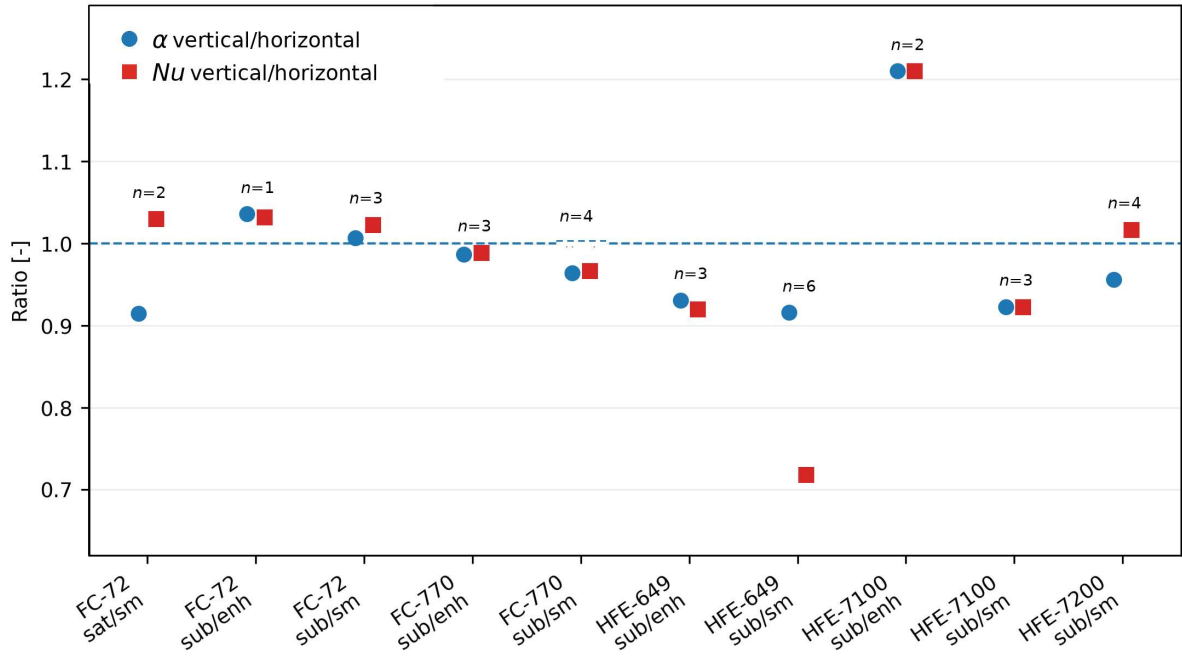


Figure 3. Vertical/horizontal orientation contrasts by working fluid, regime and heated foil surface condition. Markers show median ratios for α and Nu , n denotes the number of matched cells, and the dashed line marks parity. Abbreviations: sat = saturated, sub = subcooled, sm = smooth heated foil surface, enh = enhanced heated foil surface.

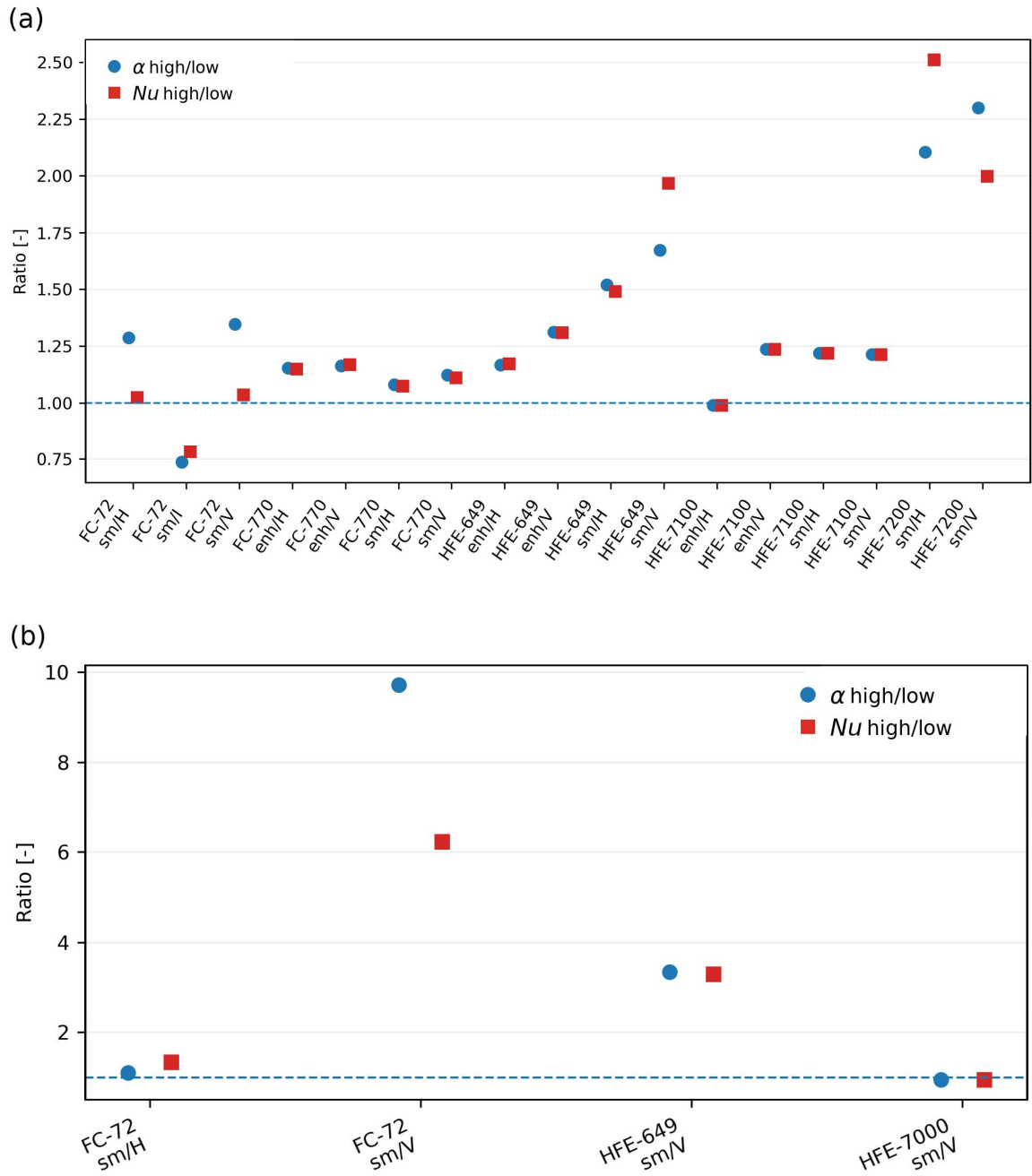


Figure 4. Nominal mass flow contrasts: (a) subcooled subsets; (b) saturated subsets. Markers show the ratios of α and Nu obtained at the highest and lowest nominal mass flow levels for each fixed combination of working fluid, regime, heated foil surface condition and orientation; the dashed line marks parity. Abbreviations: sm = smooth heated foil surface, enh = enhanced heated foil surface, H = horizontal, V = vertical, I = inclined.

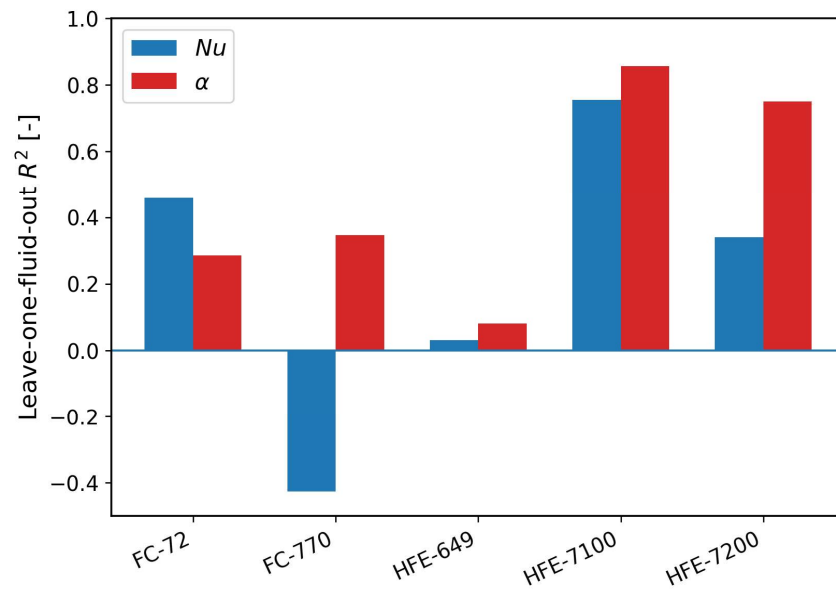


Figure 5. Leave-one-fluid-out R^2 values for the HGB model without fluid_id as a predictor.

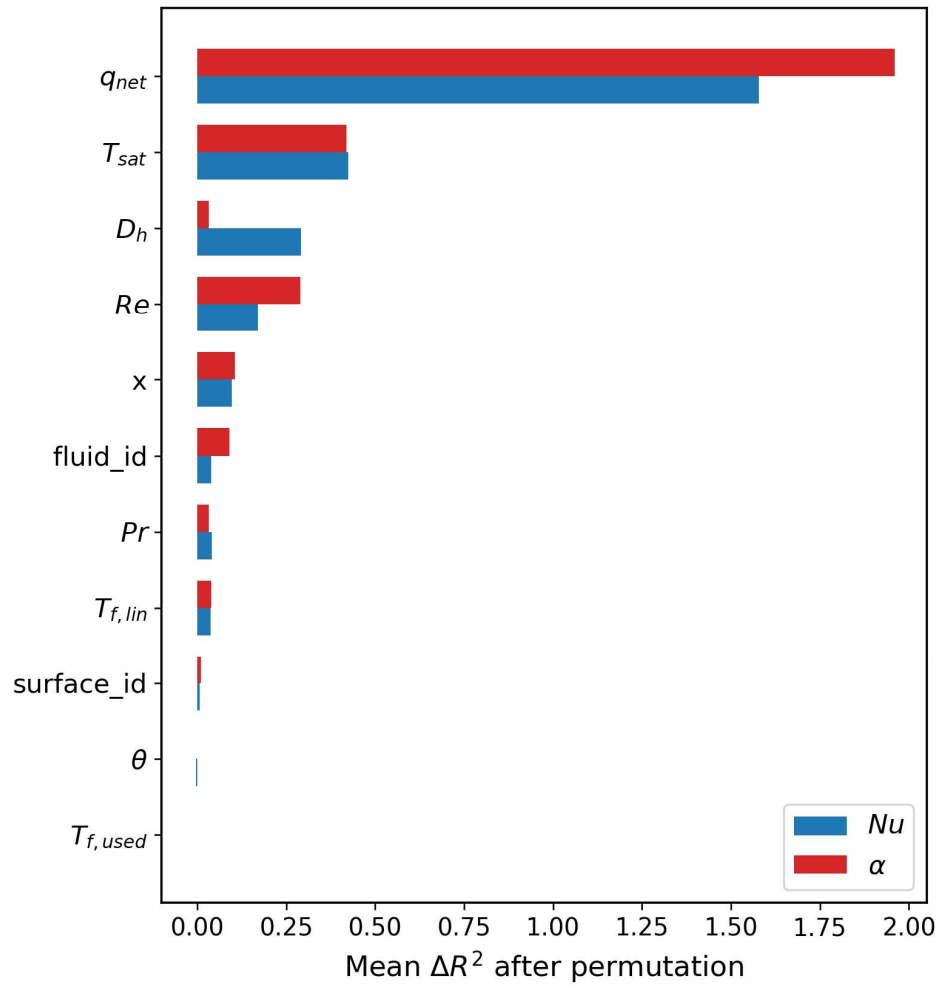


Figure 6. Permutation feature importance of individual HGB predictors, expressed as mean ΔR^2 after permutation and averaged across five repeated source-file splits after source/domain QC.